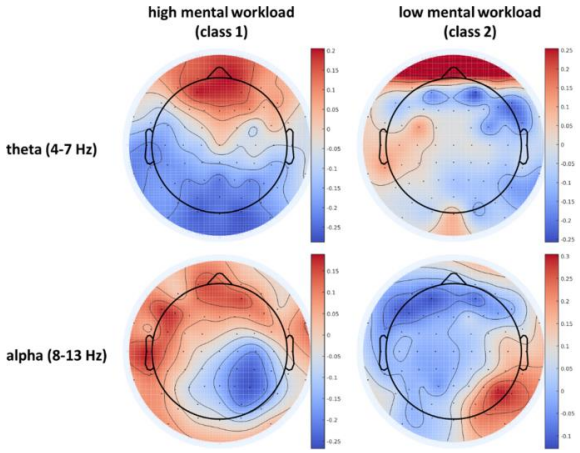
Cross-Session EEG Mental Workload Classification

1. Introduction and Problem Context

1.1 Mental Workload Monitoring

Mental workload refers to the cognitive effort an individual dedicates to a task and is one of the most studied applications of passive brain–computer interfaces because it estimates the operator's cognitive state from electroencephalography (EEG) without requiring explicit input (Gallegos Ayala et al., 2023). This matters in safety-critical multitasking environments such as surgery and process supervision, where excessive workload produces fatigue, distraction and reduced efficiency, which in turn cause errors in analysis and decision-making (Qu, Zhang and Pang, 2022).

EEG workload estimation is based on a well-known pattern: frontal theta power increases and parietal alpha power decreases as task demand rises (Pušica et al., 2025). However, while this pattern is useful for distinguishing low and high workload, it is much harder to separate the intermediate Medium class, which often overlaps with both extremes (Hassan et al., 2025). This makes three-class workload classification a more realistic and more difficult problem than binary classification.

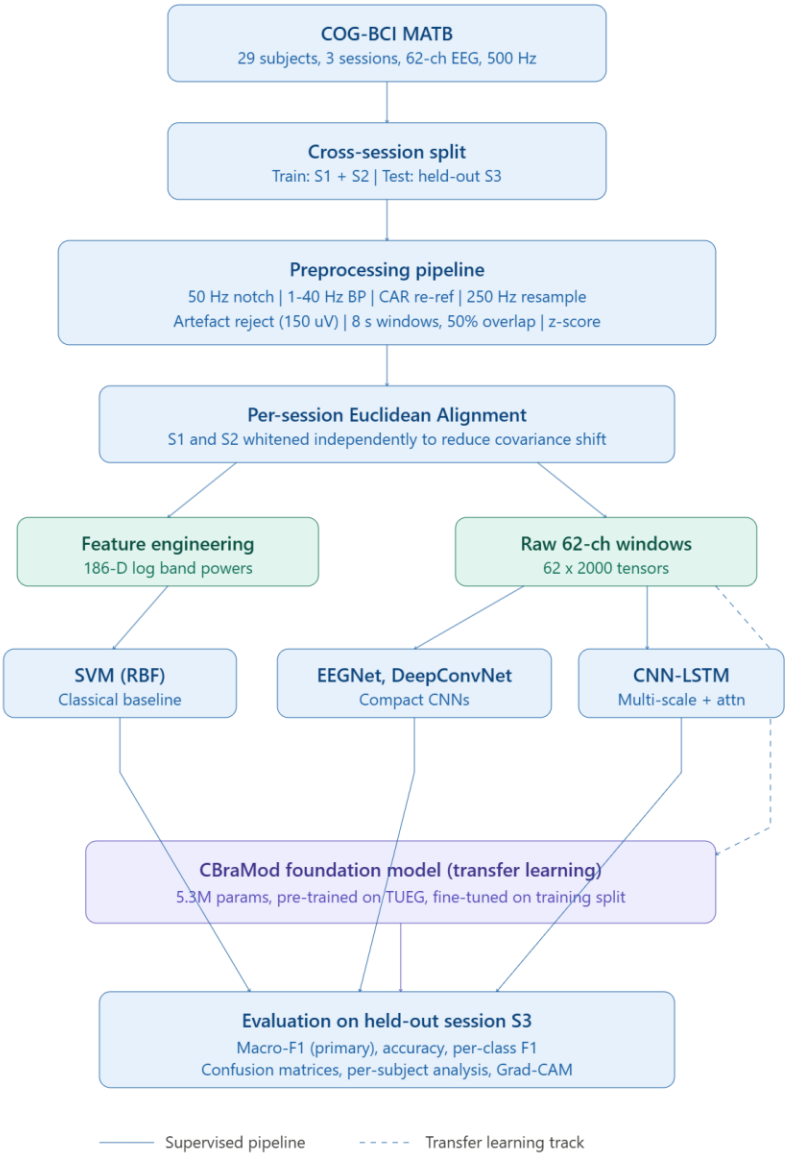


The problem is compounded by cross-session domain shift. EEG is a non-stationary signal: even when the same participant performs the same task on different days, electrode placement, impedance and cognitive baseline all change, drawing the recording from a different distribution (Qu, Zhang and Pang, 2022). A model trained on one session may therefore learn session-specific patterns rather than workload-relevant features. This makes the task not just classification but domain adaptation and means within-session accuracy is an unreliable proxy for real-world performance.

1.2 Project Aim and Scope

The problems identified have clear implications for EEG workload monitoring in practice. A classifier that does not generalise across sessions is unlikely to work reliably in real operations, and a system that cannot separate intermediate from critical demand has limited value where early intervention matters most.

The aim of this project is to identify which modelling and preprocessing choices improve cross-session workload classification, and which remain limited by session shift or class ambiguity. Rather than focusing on within-session accuracy, the project prioritises realistic evaluation and treats Medium-class performance as the main diagnostic measure. The scope is experimental rather than deployment ready. The project is intended to establish a robust baseline and identify the main technical barriers, so that future work on calibration, adaptation, and labelling can build on a validated foundation.



2. Experimental Design and Data Pipeline

2.1 Dataset

The COG-BCI database contains EEG recordings from 29 participants across three sessions and four cognitive tasks, with over 100 hours of open data (Hinss et al., 2022). It is well suited to this project because it allows evaluation of cross-session generalisation rather than simple within-session memorisation. MAT files were used because they preserved the EEG structure, simplified loading and preprocessing, and made it easier to focus on the workload-related recordings needed for classification.

Property	Value
Subjects	29 (sub-01 to sub-29; sub-05 and sub-22 contributed 0 usable windows after artefact rejection)
Sessions per subject	3 (S1, S2 → train; S3 → test)
Levels per session	3 (easy=Low, med=Medium, diff=High)
EEG channels	62 (intersection across all recordings)
Sampling rate	500 Hz raw
Total MATB recordings	261 (29 × 3 × 3)

Table 1. COG-BCI MATB dataset summary.

2.2 Preprocessing Pipeline

Although deep models can in principle operate on raw EEG, raw recordings can underperform because they retain line-noise, drift and session-varying amplitude offsets (Del Pup et al., 2025). The pipeline in Table 2 was applied in a fixed order, so filtering and artefact handling occur before scaling, and segmentation only after cleaning.

Step	What	Why
1. Notch	50 Hz zero-phase IIR	Removes UK/EU mains-line interference that would dominate beta-band features.
2. Band-pass	1–40 Hz FIR, zero-phase	Removes drift below 1 Hz and EMG above 40 Hz while preserving theta/alpha/beta.
3. Bad-channel detection	Variance z>5 SD → spherical interpolation	Repairs broken electrodes before they corrupt the common-average reference.
4. Common average reference	Subtract per-sample mean across 62 channels	Reduces reference-site artefact and stabilises across sessions.
5. Resample	500 → 250 Hz	Halves compute while keeping a generous Nyquist margin.
6. Windowing	8 s, 50% overlap → 62 × 2,000	Captures enough theta cycles for stable spectral estimates; overlap doubles the effective sample count.
7. Artefact rejection	Peak-to-peak > 150 µV → drop	Removes blinks/muscle/electrode pops; ~56% of windows discarded.
8. Per-window z-score	µ=0, σ=1 per window	Removes session-level amplitude offsets without leaking statistics across the split.

Table 2. Preprocessing pipeline applied to all recordings.

2.3 Cross-Session Split and Leakage Prevention

Sessions S1 and S2 were used for training and S3 was held out entirely for testing, with no calibration data from the test session leaking into training. This is stricter than the random splits used by 37% of studies surveyed by Pušica et al. (2025), where adjacent windows from the same session can appear in both partitions, inflating performance. Class balance after rejection is mildly imbalanced (Table 3) and was addressed with inverse-frequency class weights during training (Low=0.778, Medium=1.100, High=1.241).

Class	Train (%)	Test (%)
Low (easy)	2,397 (42.9%)	1,254 (47.5%)
Medium (med)	1,694 (30.3%)	759 (28.7%)
High (diff)	1,501 (26.9%)	627 (23.7%)
Total	5,592	2,640

Table 3. Class distribution after artefact rejection.

2.4 Domain Adaptation: Euclidean Alignment

Euclidean Alignment (EA) whitens each split using its mean covariance, reducing cross-session shift without requiring labels (He and Wu, 2020). S1 and S2 were aligned independently before concatenation to prevent one session biasing the whitening reference, producing a consistent improvement in macro-F1 across classifiers. A subsequent attempt to stack CORAL on top of EA hurt performance, confirming that adaptation methods do not stack freely when each assumes different distributional structure.

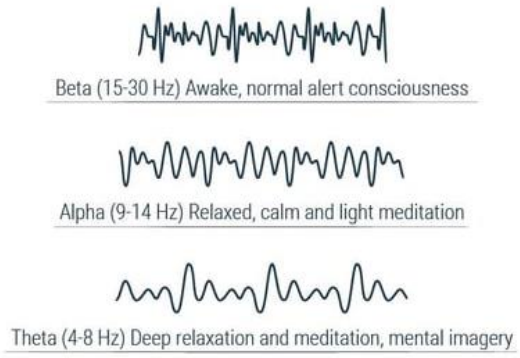
3. Machine Learning Baseline

3.1 Feature Engineering versus Representation Learning

The fundamental distinction between traditional ML and deep learning in EEG is where the inductive bias enters. Classical pipelines hand-craft features grounded in neuroscience and train a shallow classifier on top (Antony et al., 2022; Safari et al., 2024). When training data are scarce and the relevant features are well understood, hand-crafted pipelines often match deep models; once the task departs from textbook spectral templates, learned representations dominate (Lawhern et al., 2018; Hassan et al., 2025).

3.2 SVM Baseline

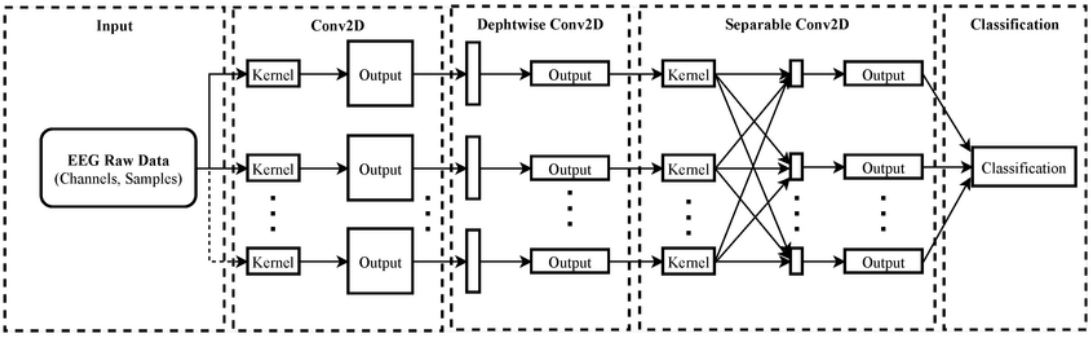
SVMs are one of the most common classical classifiers in EEG workload studies (Hassan et al., 2025). Each 8-second window was summarised using log Welch band powers in theta, alpha and beta across 62 channels, producing a 186-dimensional feature vector. An RBF kernel with balanced class weights modelled nonlinear boundaries. The baseline is intentionally strong: any gain the deep models achieve over it is evidence about representation learning, not classifier capacity.



4. Deep Learning Models

4.1 EEGNet

EEGNet is a compact CNN (~2K parameters) designed for EEG decoding, combining temporal, depthwise spatial and separable convolutions within a single architecture (Lawhern et al., 2018). It was the first deep model tried because its small size makes it suitable for limited training data, which is the case for this study with ~5,600 training windows after artefact rejection.



4.2 DeepConvNet

DeepConvNet was included as a deeper convolutional alternative to EEGNet to test whether greater architectural depth improved workload classification and to provide an intermediate comparison between EEGNet and CNN-LSTM. Unlike EEGNet, it uses stacked convolutional blocks with batch normalisation, ELU activation, max-pooling, and dropout, allowing it to learn hierarchical temporal-spatial features from EEG signals (Schirrmeister et al., 2017).

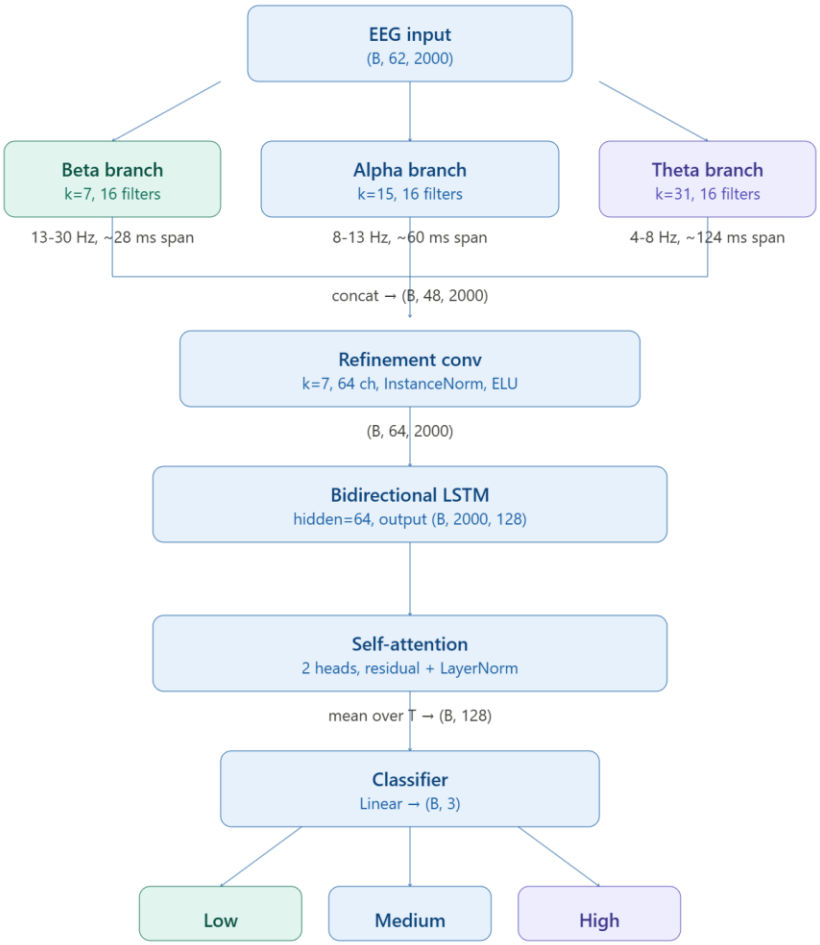
4.3 CNN-LSTM with Multi-Scale Branches and Attention

The CNN-LSTM was designed to address three specific limitations exposed by the simpler models. EEGNet's fixed temporal kernel could not simultaneously capture fast beta and slow theta activity. DeepConvNet added depth but relied on batch normalisation, whose frozen running statistics degrade under cross-session shift. Neither model had temporal memory, meaning they treated each time step independently despite workload being a sustained cognitive state. Table 5 summarises the architecture, parameter counts and the hypothesis each stage tests.

The multi-scale convolution is the most important design choice because it lets the model extract beta, alpha, and theta features in parallel instead of forcing one kernel to cover all time scales. This is especially useful for the Medium class, which combines characteristics of both Low and High workload. The refinement convolution then lets features from different bands interact, so the model can learn patterns such as high theta and low alpha at the same time.

Instance normalisation is used instead of batch normalisation because it normalises each sample separately, which helps reduce the effect of cross-session shift. The bidirectional LSTM captures temporal context across the full 8-second window, while self-attention helps the model focus on important time steps. Two attention heads were enough, and global average pooling was used because workload is treated as a sustained state rather than a momentary event.

Stage	Params	Key choices	Hypothesis tested
Beta branch Conv1d k=7	6,944	28 ms receptive field - covers one beta cycle at 250 Hz	Beta (13–30 Hz) encodes active cognitive engagement
Alpha branch Conv1d k=15	14,880	60 ms - covers fastest alpha component (~13 Hz)	Alpha suppression (8–13 Hz) marks increased load
Theta branch Conv1d k=31	30,752	124 ms - covers fastest theta (~8 Hz); largest kernel	Frontal theta (4–8 Hz) increases with workload
Refinement conv Conv1d k=7 + InstanceNorm	21,728	InstanceNorm not BatchNorm - no frozen statistics at inference	Cross-band features (e.g. high-theta + low-alpha) need one mixing layer; InstanceNorm survives session shift
BiLSTM hidden=64, 1 layer	~66,000	Bidirectional; single layer to avoid overfitting on 5,600 windows	Sustained workload requires temporal memory - conv alone cannot integrate state across 8 s
Self-attention 2 heads, embed=128	~66,000	Residual + LayerNorm; 2 heads not 8 - data scale too small for wider attention	Direct path between any two of 2,000-time steps where LSTM gradient flow is lossy
Classifier Mean pool + Linear	643	Mean-pool over time - all steps contribute equally to workload estimate	Workload is sustained; last-step readout gives undue weight to window boundary
Total	**~205K**	Upper limit of what ~5,600 training windows support without overfitting. A 700K version was tested; train macro-F1 exceeded 0.95 while test dropped below 0.45.	



4.4 Training Configuration

All deep models were trained with focal loss to concentrate gradient signal on the harder Medium class. The focusing parameter gamma was tuned per model because the architectures respond differently to hard-example weighting, gamma=2.0 improved CNN-LSTM's Medium-F1 from 0.40 to 0.51, but collapsed Low-class recall in DeepConvNet, which required gamma=1.0. EEGNet used gamma=1.5 as a compromise given its limited capacity. Label smoothing of 0.1 was applied because EEG labels are inherently noisy at the window level, since a window labelled Medium may contain transient stretches that are spectrally indistinguishable from Low or High, and overconfident predictions on ambiguous samples hurt generalisation. DeepConvNet and CNN-LSTM used moderate augmentation. Table 6 details the full CNN-LSTM hyperparameter configuration.

Hyper-parameter	Value	Why
Optimizer	AdamW (weight_decay=1e-4)	Decoupled weight decay; cleaner regularisation than Adam
Learning rate (peak)	1e-3	Standard for AdamW on this scale; higher caused divergence in epoch 1–2
LR schedule	5-epoch linear warmup → cosine annealing to 0 over 50 epochs	Warmup prevents early-epoch instability with the attention layer; cosine is monotone and dependable
Batch size	32	64 was tried, gave noisier validation curves with no accuracy benefit
Epochs	50 (no early stopping triggered)	CNN-LSTM converges late, so we let it run to LR=0
Loss	Focal loss γ=2.0 + class weights + label smoothing 0.1	γ=2 down-weights easy examples; class weights handle imbalance; label smoothing prevents over-confident predictions on noisy labels
Augmentation	Gaussian noise σ=0.05, circular shift ±50, channel dropout p=0.10	All three simulate realistic EEG nuisance variation

5. Foundation Model: CBraMod

5.1 CBraMod

CBraMod is a transformer foundation model published at ICLR 2025 (Wang et al., 2025) that uses criss-cross attention to model temporal and spatial dependencies separately rather than jointly, pre-trained on the Temple University EEG corpus through patch-based masked reconstruction. It was chosen because the project sits in exactly the regime that motivates foundation models: limited labelled data, large unlabelled clinical EEG upstream, and a need for representations that survive cross-session shift. The transferability claim rests on the substantial overlap between clinical and workload EEG: the same theta, alpha and beta rhythms dominate both, even though the cognitive content differs.

5.2 Adaptation Strategy

Two adaptation strategies were evaluated: a linear probe with the backbone frozen and only a 412K-parameter head trained, and full fine-tuning with differential learning rates (backbone 5×10^{-5}, head 1×10^{-3}). Differential rates matter because aggressive backbone updates erase the pre-training signal.

6. Evaluation and Results

6.1 Metrics Justification

Macro-F1 was chosen as the primary metric because it weights all three classes equally regardless of post-rejection imbalance, whereas accuracy is dominated by the larger Low class. Per-class F1 is reported alongside macro-F1 to make the Medium-class problem visible.

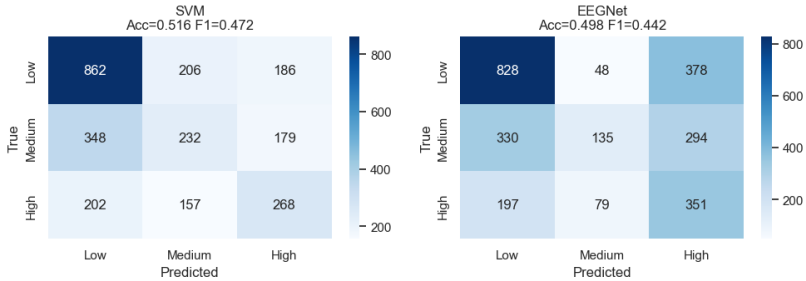
6.2 Results Comparison

Table 4 summarises cross-session performance across the five models. The progression from SVM (0.47 macro-F1) through CNN-LSTM (0.577) to CBraMod (0.625) is not a smooth ladder but a sequence of falsified hypotheses, each model revealing a specific limitation that motivated the next.

Model	Accuracy	Macro-F1	Low-F1	Medium-F1	High-F1
SVM (band-power)	0.5159	0.4716	0.6467	0.3427	0.4254
EEGNet	0.4977	0.4415	0.6347	0.2644	0.4255
DeepConvNet	0.5330	0.5232	0.6166	0.4368	0.5162
CNN-LSTM	0.5989	0.5768	0.7022	0.5128	0.5154
CBraMod fine-tuned	0.6908	0.6247	0.8400	0.5000	0.5341

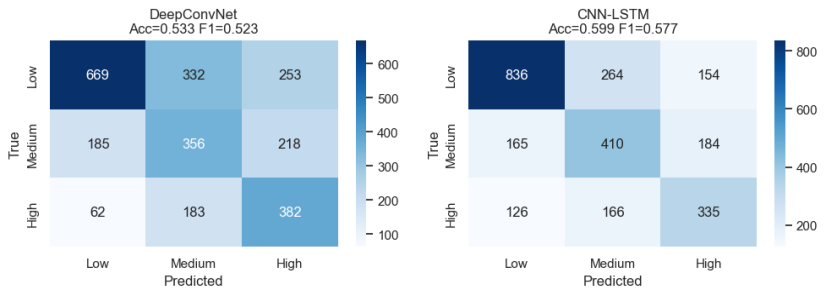
Table 4

The SVM baseline confirms that hand-crafted spectral features carry genuine workload information (0.516 accuracy), but its confusion matrix reveals a systematic bias: 348 of 759 Medium windows are misclassified as Low, indicating that log band-power features cannot reliably separate the intermediate class. This is the failure the deep models were designed to address.



EEGNet performed worse than the SVM, with 0.4977 accuracy and 0.4415 macro-F1. Its early stopping at epoch 27 suggests limited capacity under cross-session shift, making it a useful lightweight baseline but insufficient for robust three-class workload classification.

DeepConvNet improves to 0.5232 macro-F1 and lifts Medium-F1 from 0.3427 to 0.4368, confirming that depth helps. However, the gain plateaus, suggesting depth alone reaches a ceiling without frequency-specific priors or temporal memory.

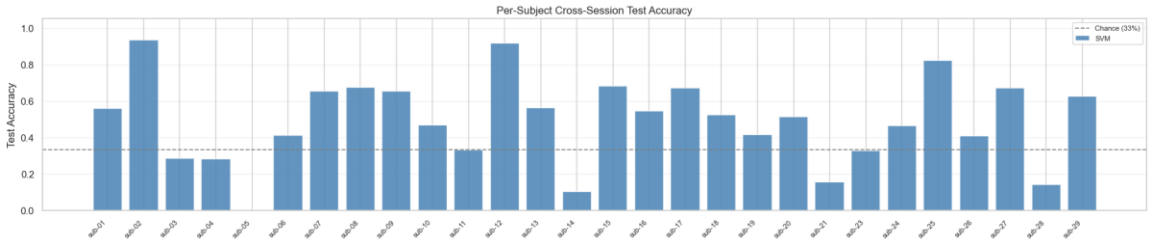


CNN-LSTM gave the strongest improvement among the from-scratch models, reaching 0.5989 macro-F1. Its Medium-F1 improved to 0.513, and Medium-to-Low errors fell sharply, showing better class separation than the SVM baseline. The smooth training curve and small train-validation gap suggest that the 205K-parameter model size was appropriate for the dataset. This supports the model design: multi-scale convolutions captured band-specific features, instance normalisation helped with session shift, and the BiLSTM added temporal context.

Fine-tuned CBraMod performed best overall, with 0.691 accuracy and 0.625 macro-F1, exceeding other models. However, it did not solve the Medium-class problem, since Medium-F1 stayed at 0.500, slightly below CNN-LSTM. Instead, it improved Low and High performance, which suggests that the Medium class is a structural challenge rather than a simple capacity issue. When the Medium class was removed entirely, binary Low-versus-High accuracy rose to 0.712, confirming that the remaining error is concentrated in the intermediate class.

6.3 Per-Subject Variance

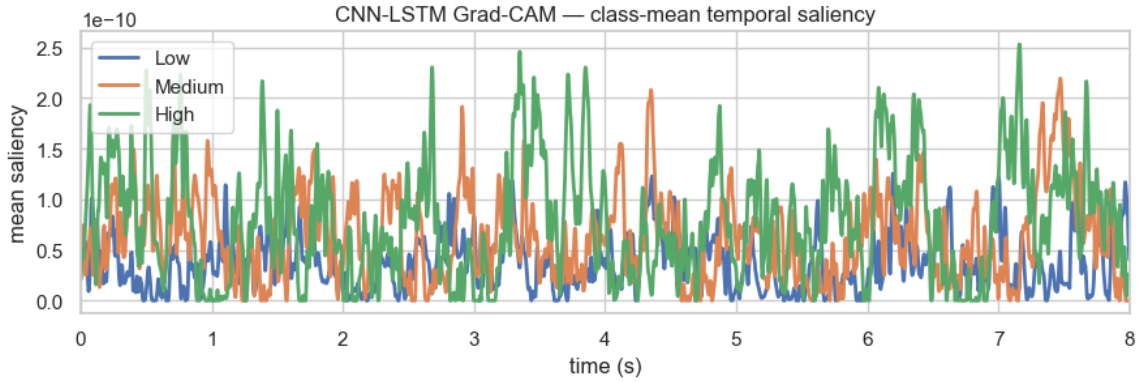
Figure below shows that the cross-session generalisation gap is not uniform but concentrated in a subgroup with stronger distribution shift. Eight of 28 subjects performed at or below chance (accuracy below 0.333), while the strongest (sub-02) reached 0.937. This is important for deployment because it means the classifier would be unreliable for some individuals, even if overall accuracy appears acceptable. Gupta et al. (2021) reported similar subject-level variation in another workload dataset and found that subject-specific models outperformed pooled models, which supports this interpretation. Although subject-specific calibration could improve performance, it would undermine the zero-calibration setting used in this project. For that reason, test-time adaptation appears to be the more promising next step.



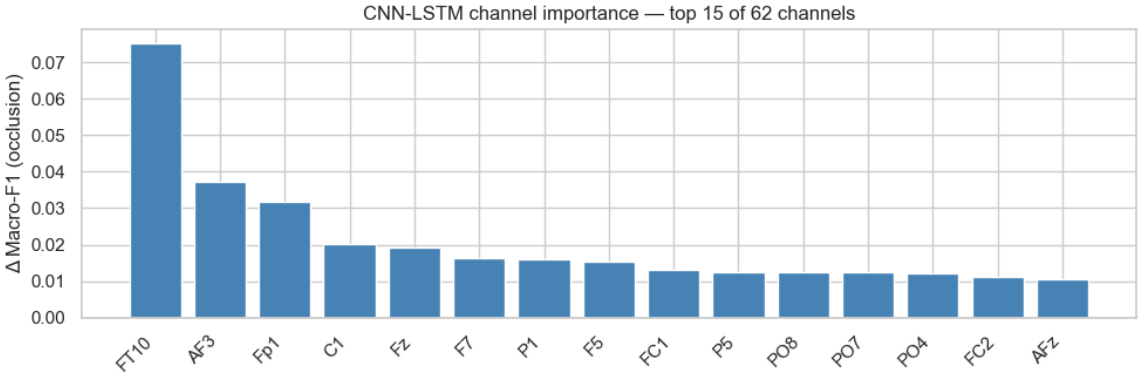
6.4 Interpretability: Grad-CAM

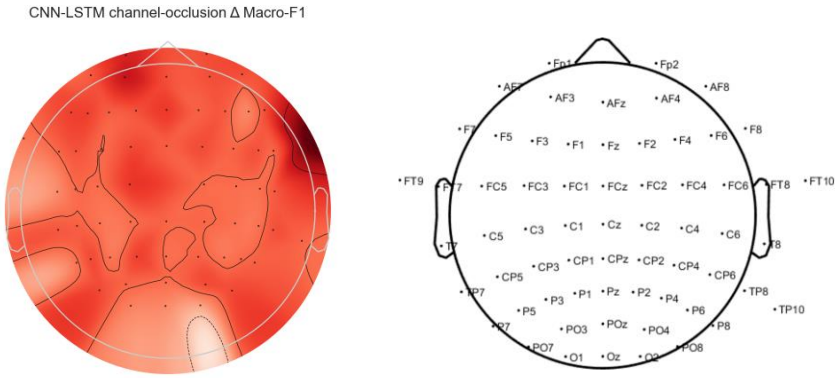
Two interpretability methods were applied to verify the CNN-LSTM is learning physiologically plausible features rather than exploiting artefacts.

Grad-CAM (Selvaraju et al., 2017) shows broadly distributed temporal saliency across the full eight-second window rather than concentration at boundaries, which would indicate artefact sensitivity. High-workload saliency is strongest, Medium-workload saliency is weaker and inconsistent (aligning with poor Medium-class F1), and Low-workload saliency is flattest, consistent with a quiet spectral state being the distinguishing feature of low demand.



Channel-occlusion analysis zeroes out one channel at a time and measures the drop in macro-F1. The most important channel is FT10 (delta F1 = 0.075), but the top 15 channels span frontal, central, parietal and temporal regions, confirming a distributed importance pattern consistent with workload being a whole-brain state. This suggests a reduced montage could retain most discriminative information, but a narrowly frontal selection would miss parietal and temporal contributions the model depends on.





7. Impact, Implications and Limitations

7.1 Impact on Individuals and Organisations

In safety-critical deployment, the main risk is asymmetric: a false negative may fail to trigger assistance when most needed, whereas a false positive typically results only in an unnecessary intervention. The eight of 28 subjects at or below chance are therefore more important than the headline accuracy, because they reveal a reliability floor below which the system would be unsafe. Cognitive autonomy is a further concern: continuous neural monitoring raises issues of consent, privacy and control over mental states that Ienca and Andorno (2017) argue warrant recognition as emerging human-rights issues. At the organisational level, deployment may shift liability, since failure to act on a system-flagged alert could be attributed to the operator.

7.2 Environmental Trade-offs

The supervised models used in this project are relatively small, with the CNN-LSTM containing approximately 205,000 parameters and requiring only minutes of training on a single GPU. By contrast, CBraMod involves a much larger pre-training cost, although this is amortised across the wider community because only fine-tuning is required for the downstream task. The greater environmental concern is continuous inference at scale, since even modest per-user power consumption becomes significant when deployed across many operators. For that reason, edge deployment with quantised models is the most responsible option, both to reduce energy use and to keep raw EEG data local.

7.3 Regulatory Context

Under the EU AI Act (European Union, 2024), biometric-state inference in employment may fall into either the prohibited or high-risk category, which brings requirements for conformity assessment, transparency, human oversight, and post-market monitoring. This makes workload monitoring not only a technical problem but also a regulatory one, since the compliance burden may be as significant as the modelling work itself. In this context, reporting per-subject reliability is especially important because it aligns with the Act’s emphasis on transparency and accountability.

7.4 Limitations

EEG's non-stationarity remains the most fundamental barrier: the signal changes across days depending on sleep, stress and electrode placement (Qu, Zhang and Pang, 2022), and the per-subject results in Section 6.3 confirm this remains unsolved. The 56% artefact rejection rate may also bias the kept data toward calmer conditions, meaning real-world performance could be worse. The Medium class remains unresolved across all models, partly because of spectral overlap and partly because MATB difficulty is set by task frequency rather than direct measurement of cognitive state, meaning two participants in the same condition may experience different levels of load. Finally, the project uses laboratory-grade 62-channel EEG, and practical deployment barriers are discussed in Section 8.

8. Emerging Trends and Future Directions

8.1 Foundation models and transfer learning

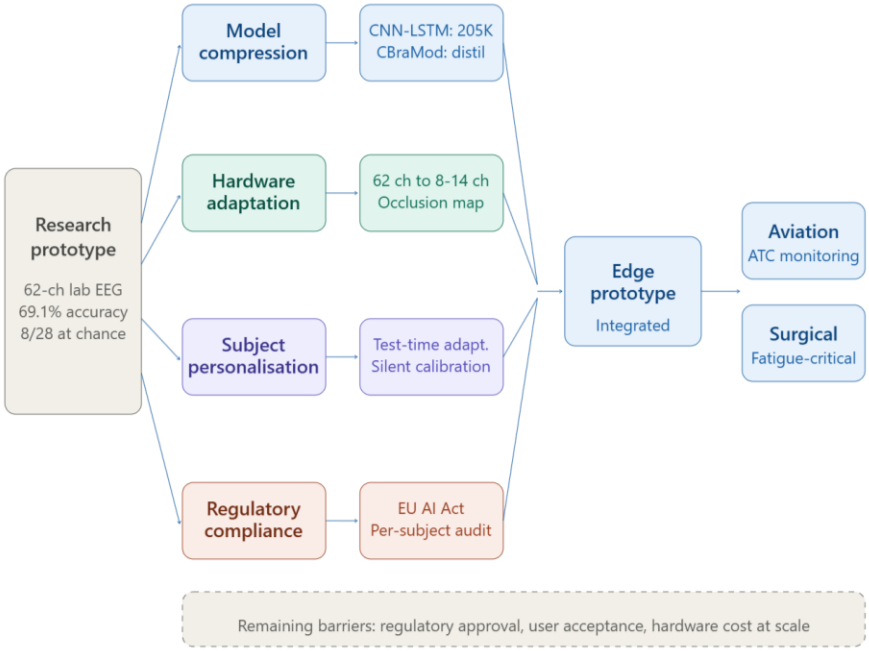
The most significant near-term development is the shift toward large pre-trained foundation models. CBraMod (Wang et al., 2025), LaBraM (Jiang et al., 2024) and BENDR (Kostas et al., 2021) all pre-train transformers on large pooled EEG corpora, then fine-tune on small labelled tasks. This project provides direct evidence for the value of this approach, since CBraMod's fine-tuned macro-F1 exceeded the best from-scratch model despite using the same training data. The practical question for future work is not whether foundation models help but how to make them deployable. Distillation or quantisation that preserves transfer ability while reducing CBraMod's 5.3M parameters to edge-viable scale would be the most impactful next step for moving this work toward real-world use.

8.2 Test-time adaptation for subject-level personalisation

The most urgent unresolved problem is the subject-level failure identified in Section 6.3. Fixed offline models cannot solve this because the issue is individual physiological variation that pre-training on other people's data cannot fully compensate for. Test-time adaptation (TTA) addresses this directly by continuing to update on unlabelled test data at inference, reducing distribution shift without new labels. This integrates naturally with the Euclidean Alignment front-end already used in this project. In deployment, TTA would mean the system silently personalises itself during the first minutes of a shift rather than requiring a calibration session.

8.3 From Research to Deployment

Laboratory EEG uses 62 gel-based electrodes with skilled setup, which is impractical in most workplaces. Consumer headsets typically provide 8 to 14 dry-electrode channels, meaning the models trained here would need channel selection or retraining. The channel-occlusion analysis in Section 6.4 provides a starting point, suggesting a targeted reduced montage could retain much of the discriminative information. The CNN-LSTM at 205K parameters is already close to edge-viable, while CBraMod would require compression. The most plausible near-term deployment scenario is niche safety-critical environments such as surgical theatres, where the cost of an overload event justifies EEG infrastructure investment.



9. Conclusion

The project tested rather than assumed the value of deep learning for cross-session EEG workload classification. A strong SVM baseline reached 0.4716 macro-F1; a custom CNN-LSTM with multi-scale convolutions, instance normalisation and attention reached 0.577; a fine-tuned CBraMod foundation model reached 0.625, exceeding the from-scratch models. The aim set out in Section 1.2, to establish a robust experimental baseline that identifies the key technical barriers, has been met. Two barriers persist: the Medium-class spectral ambiguity, which is structural rather than algorithmic, and the high-shift subject subgroup, which makes deployment unreliable for specific individuals. The most promising next steps are test-time adaptation for subject-level personalisation and model compression for edge deployment. The deeper methodological lesson is that on small, non-stationary EEG data, design choices grounded in physiology and pre-training on related corpora compound; raw parameter count alone does not.

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